Exercise 3:

Exercise 3: CASE Statements

Conditional logic

```
1. SELECT product_name, price,
   CASE
       WHEN price > 1000 THEN `Expensive'
       WHEN price BETWEEN 100 AND 1000 THEN `Mid range'
   ELSE `Budget'
   END AS price_category
   FROM products;
```

Product_name	Price	Price_category
Laptop	1200	Expensive
Phone	800	Midrange
Keyboard	45	Budget
Monitor	300	Midrange
Mouse	25	Budget

```
2. SELECT customer_name, amount,
   CASE
       WHERE WHEN amount ≥ 1000 THEN `High Value'
       WHEN amount BETWEEN 500 AND 999.99 THEN `Medium Value'
       WHEN amount <500 THEN `Low value'
   END AS Order_value_category
   FROM orders;
```

Customer_name	Amount	Order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

3. SELECT Emp_name, department, salary,
CASE
~~WHEN salary > 80000 THEN~~
WHEN department IN ('IT') AND salary > 80000 THEN 'Senior IT'
WHEN department IN ('HR') AND salary > 55000 THEN 'Experienced HR'
ELSE 'Staff'
END AS position_level
FROM Employees;

Emp_name	Department	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. SELECT student_name, score,
CASE
WHEN score ≥ 90 THEN 'A'
WHEN score BETWEEN 80 AND 89 THEN 'B'
WHEN score BETWEEN 70 AND 79 THEN 'C'
WHEN score BETWEEN 60 AND 69 THEN 'D'
WHEN score ~~D~~ < 60 THEN 'F'
END AS grade
FROM students;

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. SELECT delivery_Id, delivery_time_minutes,
CASE
WHEN delivery_time_minutes ≤ 30 THEN `Fast'
WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN `On Time'
~~WHEN delivery_time_minutes > 60 THEN `Late~~ ELSE `Late'
END AS performance
FROM ~~Tickets~~ Deliveries;

Delivery_Id	Delivery_time_minutes	Performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	~~[illegible]~~ Late

6. SELECT Ticket_Id, Issue_type, Priority,
CASE
WHEN Priority = 3 THEN `High',
WHEN Priority = 2 THEN `Medium'
WHEN Priority = 1 THEN `Low'
END AS Priority_label
FROM Tickets;

Ticket_Id	Issue_type	Priority	Priority_label
1	Login. Issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email down	2	Medium
5	Password reset	1	Low

~~7 SELECT student_Id, attendance percentage, attendance~~

7. ~~SELECT student_Id, SUM(days_present)/~~

```
7. SELECT student_Id, SUM(days_present)* 100 /SUM(total_days)
   AS Attendance_percentage,
CASE
    WHEN Attendance_percentage ≥ 90 THEN 'Excellent'
    WHEN Attendance_percentage BETWEEN 75 AND 89 THEN 'Good'
    WHEN Attendance_percentage <75% THEN 'Needs Improvement'
   END AS Attendance_status
FROM Attendance;
```

Product_Id	Attendance_Percentage	Attendance_Status
1	90%	Excellent
2	65%	Needs Improvement
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

```
8. SELECT product_Id, stock_qty,
CASE
   WHEN Stock_qty = 0 THEN 'Out_of_stock'
   WHEN Stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
   ELSE 'IN Stock'
END AS Stock_status
FROM Products_inventory
```

Product_Id	Stock_qty	Stock_status
1	5	In Stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

```
9. SELECT subject, enrolled_~~category~~ students,
CASE
    WHEN Enrolled_students ≥ 25 THEN 'Large'
    WHEN Enrolled_students, BETWEEN 10 AND 24 THEN 'Medium'
    WHEN Enrolled_students < 10 THEN 'Small'
END AS class_size_category
FROM classes
```

~~Class_Id~~ Subject	Enrolled_students	Class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

```
10. SELECT payment_Id, amount, payment_method,
CASE
    WHEN payment_method = 'Cash' AND amount ≥ 200 THEN
    'Eligible for discount'
ELSE 'Not Eligible'
END AS Discount_eligibility
FROM payments;
```

Payment_Id	Amount	Payment_method	Discount_eligibility
1	50.00	Card	Not eligible
2	200.00	Cash	Eligible for discount
3	150.00	Card	Not eligible
4	75.00	PayPal	Not eligible
5	300.00	Cash	Eligible for discount